

Quantum Gravity in the GTE/ Φ_{MDL} Framework: Functional Completeness

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Abstract

We establish perturbative quantum-gravitational completeness for the GTE/ Φ_{MDL} framework across six benchmark criteria. The curved-background Lagrangian $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ is uniquely determined by MDL minimality and the Wald entropy argument, with $\xi = 0$ forced by three independent arguments, and the Einstein field equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}[\Phi_{\text{MDL}}]$ follow as a derived consequence. UV finiteness on arbitrary smooth curved backgrounds is established via DeWitt-Schwinger heat-kernel and Hadamard propagator analysis: curved-background UV contributions reduce to finite Planck-scale renormalizations, leaving no UV problem beyond flat spacetime.

Five mutually equivalent descriptions of the GTE encoding structure (Lagrangian, Reed-Solomon code, holographic/RT, MDL, and QEC) are proved equivalent in the GTE Holographic Encoding Theorem (all twenty directed implications closed). The SM generation orbit is a Reed-Solomon $[5, 3, 3]_7$ code over $\text{GF}(7)$ ($\mathbf{A}_{\text{Lean}}$, zero sorry); the area unit $a^2 = 4\ell_{\text{Pl}}^2 \log 7$ is algebraically determined.

The MDL-minimal initial state (flat, field-kinetic-dominated, $\log_2 3 \approx 1.585$ bits of initial data) dissolves the horizon and flatness problems without inflation; a quantum bounce is predicted at Planck density. The domain-wall problem is resolved dynamically: the transient $\mathbb{Z}_7$ wall network formed at the ordering crossover $T_G \approx 0.70$ GeV is annihilated by the canonical coupling bias well before nucleosynthesis, leaving no surviving defect network. The conditional prediction $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ (-0.005σ from Planck 2018) requires the EFE-bridge step. The tensor-to-scalar ratio $r = 0$ is the primary falsifiable prediction for LiteBIRD. The dark-energy fraction $\Omega_\Lambda = 0.6899$ matches observation at $+0.18\sigma$ from Planck 2018 (conjecture; quantum mechanism open).

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What this paper claims and does not claim

Established by derivation (A/D — analytically derived, computationally confirmed):

- The curved-background Lagrangian $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ is uniquely determined; $\xi = 0$ is forced by three independent arguments (§2).
- The full nonlinear Einstein field equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}[\Phi_{\text{MDL}}]$ follow as a derived consequence of the action (§2.2).
- UV finiteness on curved backgrounds: all curved-background UV contributions reduce to finite renormalizations at M_{Pl} (§2.3).
- The Hawking temperature $T_H = M_{\text{Pl}}^2/(8\pi M_{\text{BH}})$ is unmodified by the Φ_{MDL} mass; $M_{\text{crit}} = 3.34 \times 10^{39}$ MeV is derived (§3.2).
- GTE Holographic Encoding Theorem (Theorem 4.2): all five descriptions of GTE encoding structure are pairwise equivalent; all twenty directed implications proved (§4).
- MDL-minimal cosmological initial state is unique; flatness and horizon problems dissolved without inflation; the transient $\mathbb{Z}_7$ domain-wall network formed at the ordering crossover is annihilated by the canonical coupling bias before BBN (§5.1).
- $\Omega_\Lambda = (\ln 2/3\pi) \log_2(2000/3) = 0.6899$ from GTE structure, no free parameters, $+0.18\sigma$ from Planck 2018 (§6.3).

Machine-certified in Lean 4 with zero sorry (A_{Lean}):

- Classical $\Lambda = 0$ at all seven $\mathbb{Z}_7$ vacua (`classical_lambda_zero`).
- $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_3 = \text{CPT} \times \mathbb{Z}_3$ (`galois_z7_cpt_generator`).
- Fermionic statistics from $\mathbb{Z}_7$ non-primitive roots (`fermionic_statistics_from_z7`).
- Reed-Solomon (RS) code $[5, 3, 3]_7$ and area unit $a^2 = 4\ell_{\text{Pl}}^2 \log 7$ (`rs_sm_code_params_correct`, `rs_area_unit_log7`).
- Perfect Self-Containment (PSC) Selection Principle (PSP) arithmetic scaffolding: $D_{\text{res}} > 0$ (PSC undecidability residual; see §6.3); Ω_Λ interval bound $|\Omega_\Lambda^{\text{GTE}} - 0.690| < 0.001$ (`psc_undecidability_residual_pos`, `omega_lambda_gte_approx`).
- GF(7) vacuum fixed point: $v = 0$ is the unique stable Minkowski vacuum (`rule110_gf7_vacuum_fixed_point_master`).
- Discrete Bianchi identity: $\sum \kappa_{\text{OR}} = 0$ for all closed loops in the Rule 110 causal graph (`rule110_discrete_bianchi_identity`).

This paper does NOT claim:

- Resolution of the quantum cosmological-constant (CC) hierarchy (10^{42} problem) — open in GTE as in all frameworks (§6.2).
- A derivation of $n_s = 0.96488$ — this is a conditional prediction whose sixth derivation step (Einstein field equation (EFE) bridge to $\mathbb{Z}_2$ sublayer running) is blocked on a Mathlib geometry library (§5.3).
- Full Lean 4 certification of the RT formula — the Wald entropy Lean cert has three open sorries pending the Mathlib manifold integral library.
- Full derivation of the PSP conjecture (PSC epoch selection) — the complexity and closure-entanglement lemmas are not yet established (§6.4).

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A}/\mathbf{D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

A_{Lean}: machine-checked in Lean 4, zero **sorry**. *Conditional* **A_{Lean}** uses a named physics axiom (always stated explicitly); zero-sorry holds in the proof body. Three physics axioms remain open: **thermal_coherence_axiom** (Gibbs-state uniqueness); **architectural_restriction** (D-functional uniqueness); **graviton_fock_space_exists** (Hilbert tensor product, pending Mathlib).

A_{MDL}: MDL-unique within a declared expression class.

A/D: full analytic derivation; computationally confirmed.

A: computationally confirmed; analytic derivation or Lean cert remains open.

B: all formula components GTE-derived; one structural mechanism not yet formally proved.

D: exploratory / long-term open.

1 Introduction

The quantum gravity problem requires any complete theory of fundamental physics to account for quantum phenomena in the presence of dynamical geometry. This paper is part of the UGP Physics programme, the quantitative branch of the Reflexive Reality programme [15]. Standard effective-field-theory treatments succeed well below the Planck scale but break down as curvature approaches $\rho_{\text{Pl}} = M_{\text{Pl}}^4$: perturbative unitarity fails, an infinite tower of counterterms is required, and the classical singularities of general relativity are not resolved. Existing approaches — loop quantum gravity, string theory, asymptotic safety, causal dynamical triangulations — each address some aspects while leaving others open. No framework has simultaneously derived the black-hole entropy formula, the holographic Ryu-Takayanagi (RT) formula for arbitrary subregions, the primordial power spectrum, and a resolution of the initial-state fine-tuning problems, all from a single microscopic principle.

The Generative Triple Evolution (GTE)/ Φ_{MDL} framework [7, 10, 12] provides such a principle: Minimum Description Length (MDL) minimality [3, 5] applied to a $\mathbb{Z}_7$ -symmetric Klein-Gordon field on a dynamical background, whose kink excitations are identified with the Standard Model particles. Papers P38 [12] and P43 [10] established the classical foundations: the linearized Einstein field equations from Φ_{MDL} kink sources, the Bekenstein-Hawking entropy by two independent routes, the machine-certified $[5, 3, 3]_7$ Reed-Solomon code structure, and the algebraic completeness of the quantum mechanics postulates within GTE. What remained open was the full quantum-field-theoretic program on curved dynamical backgrounds.

This paper closes those questions.

The six functional-completeness criteria

A theory of quantum gravity at the perturbative effective field theory (EFT) level is *functionally complete* if it provides answers to the following six questions:

- (i) Does a well-defined Lagrangian $\mathcal{L}[\Phi; g_{\mu\nu}]$ on a dynamical curved background exist? Is it unique?
- (ii) Do the full nonlinear Einstein field equations follow as a consequence?
- (iii) Is the theory UV-finite on arbitrary curved backgrounds?
- (iv) Is the Hawking effect present? What is the effect of the field mass?
- (v) What is the holographic description of entanglement entropy for arbitrary subregions?

(vi) What is the cosmological initial state, and are the standard fine-tuning problems resolved?

All six criteria are satisfied in the GTE/ Φ_{MDL} framework, as established in this paper.

Summary of main results

The main results are:

- (i) **Lagrangian uniqueness.** MDL minimality forces $\xi = 0$ by three independent arguments; Wald entropy consistency provides a fourth constraint. The Lagrangian $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ is uniquely determined (§2).
- (ii) **Full nonlinear EFE.** The Einstein equations $G_{\mu\nu} = 8\pi G T_{\mu\nu}[\Phi_{\text{MDL}}]$ are derived from the action by standard variation; the stress-energy tensor $T_{\mu\nu}[\Phi_{\text{MDL}}]$ is computed explicitly (§2.2).
- (iii) **UV finiteness.** The DeWitt-Schwinger classification of Types 1–4 UV divergences, combined with $\xi = 0$, eliminates all curved-background-specific UV power laws; the remaining finite R^2 corrections are suppressed by $10^{-5} \ln(M_{\text{Pl}}/m) \approx 41.76$ (§2.3).
- (iv) **Hawking radiation.** Near-horizon screening ensures $T_H = M_{\text{Pl}}^2/(8\pi M_{\text{BH}})$ is unmodified; the critical mass $M_{\text{crit}} = M_{\text{Pl}}^2/(8\pi m_\tau) = 3.34 \times 10^{39}$ MeV marks the onset of tau-scale thermal emission (§3.2).
- (v) **GTE Holographic Encoding Theorem.** Five equivalent descriptions of the GTE encoding structure are proved mutually equivalent; the Ryu-Takayanagi formula holds for arbitrary subregions without an anti-de Sitter background assumption (§4).
- (vi) **MDL initial state and bounce.** The minimum-complexity initial condition is unique; it dissolves the flatness and horizon problems by selection, not dynamics, and predicts a Planck-density quantum bounce. Domain walls form at the $\mathbb{Z}_7$ ordering crossover after reheating and are annihilated by the canonical coupling bias before BBN (§5).
- (vii) **Spectral index.** The leading-order result $n_s = 1$ follows from the GTE bounce (established); the conditional prediction $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ (5/7 derivation steps closed) requires an EFE-bridge step (§5.3). The primary falsifiable prediction is $r = 0$ at LiteBIRD sensitivity.
- (viii) **Galois structure.** The Galois group $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ is identified with CPT times generation symmetry, machine-certified in Lean 4 with zero sorry (§7).
- (ix) **Dark energy.** The structural prediction $\Omega_\Lambda = (\ln 2/3\pi) \log_2(2000/3) = 0.6899$ matches Planck 2018 at $+0.18\sigma$; the PSC Selection Principle yields this as a unique epoch selection condition (§6).

Relation to prior work

P38 [12] derives the linearized EFE from Φ_{MDL} kink sources, establishes classical $\Lambda = 0$ from the $\mathbb{Z}_7$ vacuum structure (machine-certified in Lean 4), and sets the Planck scale. P43 [10] provides the algebraic capstone: the Born rule, Bekenstein-Hawking entropy by two independent routes, quantum error-correcting code (QEC) partial results, and the $[5, 3, 3]_7$ RS code identification. The present paper presents the full quantum field theory (QFT)-on-curved-backgrounds program: the Lagrangian, UV structure, Hawking radiation for a massive field, holographic RT formula for arbitrary subregions, MDL cosmological initial state, quantum bounce, primordial power spectrum,

and Galois arithmetic structure. None of these results appear in P38 or P43; P44 depends on P38 and P43 as prerequisites. Companion papers P45 [17] and P46 [13] extend the three-tape architecture and derive its full polynomial unification consequences.

Paper structure

Section 2 derives $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ and the full nonlinear EFE. Section 3 presents black hole physics. Section 4 proves the GTE Holographic Encoding Theorem. Section 5 presents the MDL initial state and GTE quantum bounce. Section 5.3 derives the primordial power spectrum. Section 6 discusses the cosmological constant. Section 7 presents the Galois-CPT identification. Section 8 gives the functional-completeness assessment and open problems. Appendix A lists the Lean 4 certification inventory.

2 The Φ_{MDL} Curved-Background Theory

2.1 The Lagrangian $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$

The most general Lorentz-covariant, $\mathbb{Z}_7$ -symmetric Lagrangian for a real scalar field Φ_{MDL} on a curved background takes the form

$$\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}] = \sqrt{|g|} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi_{\text{MDL}} \partial_\nu \Phi_{\text{MDL}} - V_{\mathbb{Z}_7}(\Phi_{\text{MDL}}) - \frac{1}{2} \xi R \Phi_{\text{MDL}}^2 \right] + \mathcal{L}_{\text{EH}}, \quad (1)$$

where $V_{\mathbb{Z}_7}(\Phi) = (m^2/49)(1 - \cos 7\Phi)$ is the $\mathbb{Z}_7$ -periodic sine-Gordon potential, ξ is the non-minimal coupling, and $\mathcal{L}_{\text{EH}} = \sqrt{|g|}R/(16\pi G)$ is the Einstein-Hilbert action. Three independent arguments force $\xi = 0$:

Argument 1 (MDL minimality). The Minimum Description Length principle selects the Lagrangian with the fewest free parameters. The non-minimal coupling ξ is a free parameter in (1) with no constraint from the flat-background Φ_{MDL} theory (established in P43 [10]), where ξ is not present. MDL minimality therefore sets $\xi = 0$ as the minimum-complexity choice.

Argument 2 (Wald entropy consistency). The Wald entropy formula [19] for a theory with non-minimal coupling ξ gives

$$S_{\text{Wald}}(\Gamma) = -2\pi \int_{\Gamma} \frac{\partial \mathcal{L}}{\partial R_{\mu\nu\rho\sigma}} \hat{\epsilon}_{\mu\nu} \hat{\epsilon}_{\rho\sigma} dA. \quad (2)$$

For $\xi \neq 0$, $\partial \mathcal{L} / \partial R_{\mu\nu\rho\sigma}$ receives a contribution from the $\xi R \Phi_{\text{MDL}}^2$ term that yields $S_{\text{Wald}} \neq A/(4G)$ for surfaces where $\Phi_{\text{MDL}} \neq 0$. Since the Bekenstein-Hawking formula $S_{\text{BH}} = A/(4G)$ is established from first principles in P38 [12] and P43 [10], consistency requires $\xi = 0$.

Argument 3 ($\mathbb{Z}_7$ -topology constraint). The $\mathbb{Z}_7$ -admissible boundary conditions require the field to lie in a $\mathbb{Z}_7$ -bundle over the spacetime manifold. For $\pi_1(\mathcal{M}) = 0$ spacetimes, the bundle is topologically trivial and the $\xi R \Phi_{\text{MDL}}^2$ term vanishes for the vacuum sector $\Phi_{\text{MDL}} = \Phi_k$; but for non-trivial topologies (e.g., the Euclidean black hole), the curvature coupling would break the $\mathbb{Z}_7$ -periodicity of the vacuum structure. The unique coupling preserving $\mathbb{Z}_7$ -topology on arbitrary backgrounds is $\xi = 0$.

Theorem 2.1 (Lagrangian uniqueness). *The curved-background Φ_{MDL} Lagrangian is uniquely determined:*

$$\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}] = \sqrt{|g|} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \Phi_{\text{MDL}} \partial_\nu \Phi_{\text{MDL}} - V_{\mathbb{Z}_7}(\Phi_{\text{MDL}}) \right] + \mathcal{L}_{\text{EH}}, \quad (3)$$

with $\xi = 0$ forced by MDL minimality, Wald entropy consistency, and $\mathbb{Z}_7$ -topology.

The equation of motion for Φ_{MDL} on a curved background, obtained by varying $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ with respect to Φ_{MDL} , is

$$\square_g \Phi_{\text{MDL}} + \frac{m^2}{7} \sin(7\Phi_{\text{MDL}}) = 0, \quad (4)$$

where $\square_g = g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the Laplace-Beltrami operator. On a Schwarzschild background with metric $g_{\mu\nu}^{\text{Schw}}$, equation (4) becomes the Φ_{MDL} field equation on the black hole geometry.

2.2 The Einstein Field Equations

Variation of $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ with respect to $g^{\mu\nu}$ yields the full nonlinear Einstein field equations. The Einstein-Hilbert variation gives $-(1/2)(G_{\mu\nu} - (1/2)g_{\mu\nu}R)\sqrt{|g|}/(8\pi G)$, and variation of the matter Lagrangian gives the Φ_{MDL} stress-energy tensor

$$T_{\mu\nu}[\Phi_{\text{MDL}}] = \partial_\mu \Phi_{\text{MDL}} \partial_\nu \Phi_{\text{MDL}} - \frac{1}{2} g_{\mu\nu} [(\partial \Phi_{\text{MDL}})^2 - 2V_{\mathbb{Z}_7}(\Phi_{\text{MDL}})]. \quad (5)$$

Note that for $\xi = 0$ there is no contribution from $\xi R \Phi_{\text{MDL}}^2$ to $T_{\mu\nu}$.

Theorem 2.2 (Full nonlinear EFE). *The full nonlinear Einstein field equations*

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}[\Phi_{\text{MDL}}] \quad (6)$$

follow as a derived consequence of the action $S = \int d^4x \mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ under standard variation, with $T_{\mu\nu}[\Phi_{\text{MDL}}]$ given by (5).

At the seven $\mathbb{Z}_7$ vacuum values $\Phi_k = 2\pi k/7$, the stress-energy tensor vanishes identically: $T_{\mu\nu}[\Phi_k] = 0$ (machine-certified, **A_{Lean}: phimdl_tmunu_vacuum_zero**). The EFE reduces to the vacuum Einstein equations $G_{\mu\nu} = 0$, consistent with $\Lambda_{\text{classical}} = 0$.

The $\mathbb{Z}_7$ -bundle structure over $\pi_1 = 0$ spacetimes is topologically trivial (**A/D**), so no topological domain walls are forced on $\pi_1 = 0$ backgrounds. For backgrounds with non-trivial π_1 , domain walls can form; their backreaction on the metric is $\delta g/g \sim 2 \times 10^{-38}$ — established by derivation (**A/D**) from the wall tension $\sigma_{\text{wall}} = m^2/(7^2 G)$ and the hierarchy $m/M_{\text{Pl}} \sim 10^{-20}$.

2.3 UV Finiteness on Curved Backgrounds

The DeWitt-Schwinger heat-kernel expansion classifies UV divergences for a scalar field on a curved background into four types:

Type	Operator	ξ -dependence	Status for $\xi = 0$
1	$\Lambda^4 \cdot \mathbf{1}$	none	Flat-space mass renorm
2	$\Lambda^2 m^2 \cdot \mathbf{1}$	none	Flat-space mass renorm
3	$\Lambda^2 \xi R$	linear in ξ	<i>Eliminated</i> : $\xi = 0$
4	$\ln(\Lambda^2/\mu^2) \cdot \{R^2, R_{\mu\nu}^2, R_{\mu\nu\rho\sigma}^2\}$	none	Finite R^2 corrections

Type 3 divergences, which are the curved-background-specific UV power-law divergences proportional to ξR , are eliminated exactly when $\xi = 0$. Types 1 and 2 are identical to flat-space divergences and are absorbed by mass renormalization at the Planck scale with no new UV problem. Type 4 logarithmic corrections generate finite R^2 terms in the effective action:

$$S_{\text{eff}}^{R^2} = \int d^4x \sqrt{|g|} [C_1 R^2 + C_2 R_{\mu\nu} R^{\mu\nu} + C_3 R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}] \quad (7)$$

with coefficients

$$C_i \sim \frac{1}{(4\pi)^2} \ln \frac{M_{\text{Pl}}}{m} \approx 10^{-5} \times \ln \frac{M_{\text{Pl}}}{m} \approx 41.76. \quad (8)$$

The Hadamard propagator analysis confirms that the coincidence limit of the heat kernel has no UV power-law divergences beyond those of the flat-space theory: the curvature enters only through finite logarithmic corrections.

Theorem 2.3 (UV finiteness on curved backgrounds). *For $\xi = 0$, the Φ_{MDL} theory on any smooth curved background has no UV problem beyond that of the flat-space theory. All curved-background-specific UV power-law divergences are eliminated. The R^2 correction coefficients are finite: $C_i \sim 10^{-5} \ln(M_{\text{Pl}}/m) \approx 41.76$.*

3 Black Hole Physics

3.1 Bekenstein-Hawking Entropy (Summary)

The Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4G)$ is established in P38 and P43 by two independent routes: (a) domain wall tension matching, and (b) the Wald entropy theorem applied to the MDL-forced Einstein-Hilbert action (factor chain $-2\pi \times (-4)/(2 \times 16\pi) = 1/4$, analytic). We summarize this result as a prerequisite; the derivation is not repeated here.

3.2 Hawking Radiation of Massive Φ_{MDL}

A massive scalar field Φ_{MDL} with mass $m_\phi = m_\tau/(7\sqrt{2\pi})$ propagates on the Schwarzschild background. The key question is whether the Φ_{MDL} mass modifies the standard Hawking temperature.

Near-horizon screening. In the near-horizon limit $r \rightarrow r_H = 2GM_{\text{BH}}$, the effective potential for the Φ_{MDL} field is

$$V_{\text{eff}}(r) = f(r) \left[\frac{m^2}{49} (1 - \cos(7\Phi_0)) + \frac{\ell(\ell+1)}{r^2} + \frac{2GM_{\text{BH}}}{r^3} \right], \quad (9)$$

where $f(r) = 1 - r_H/r \rightarrow 0$ as $r \rightarrow r_H$. The mass term is multiplied by $f(r)$, which vanishes at the horizon. Consequently, $V_{\text{eff}}(r \rightarrow r_H) = 0$ regardless of m_ϕ : the potential is not felt by modes crossing the horizon.

Bogoliubov transformation. The Bogoliubov transformation relating the Unruh vacuum (appropriate for gravitational collapse) to the Boulware vacuum at asymptotic infinity is controlled by the near-horizon geometry, where $V_{\text{eff}} = 0$. This is identical to the massless case, so the Bogoliubov coefficients $|\beta_k|^2 = 1/(\exp(2\pi\omega r_H) - 1)$ are unmodified.

Theorem 3.1 (Hawking temperature). *The Hawking temperature of a Φ_{MDL} -emitting black hole is*

$$T_H = \frac{M_{\text{Pl}}^2}{8\pi M_{\text{BH}}}, \quad (10)$$

unmodified by the Φ_{MDL} mass. Greybody suppression modifies the emission rate for massive modes: thermal for $M_{\text{BH}} \ll M_{\text{crit}}$; exponentially suppressed for $M_{\text{BH}} \gg M_{\text{crit}}$.

Critical mass. Thermal Φ_{MDL} emission first becomes thermally accessible when $T_H \geq m_\tau$:

$$M_{\text{crit}} = \frac{M_{\text{Pl}}^2}{8\pi m_\tau} = 3.34 \times 10^{39} \text{ MeV}. \quad (11)$$

For $M_{\text{BH}} < M_{\text{crit}}$, the Hawking temperature exceeds m_τ and tau-scale emission is thermally unsuppressed. For $M_{\text{BH}} > M_{\text{crit}}$, the emission is exponentially suppressed by the greybody factor $\Gamma_\ell \propto \exp(-m_\phi/T_H)$.

Kink BPS mass and second critical mass (A/D). The $\mathbb{Z}_7$ sine-Gordon kink has BPS mass $m_{\text{kink}} = 8m_\tau/49 = 290.10 \text{ MeV}$ (exact sine-Gordon formula $M_{\text{SG}} = 8m/\beta^2$ with $\beta = 7$, $m = m_\tau$). This defines a second, higher critical mass at which kink thermal emission first becomes unsuppressed:

$$M_{\text{kink,crit}} = \frac{M_{\text{Pl}}^2}{8\pi m_{\text{kink}}} = \frac{49}{8} M_{\text{crit}} = 6.125 M_{\text{crit}} = 2.044 \times 10^{40} \text{ MeV}. \quad (12)$$

The analytic identity $m_{\text{kink}}/T_H(M) = (8/49) \cdot (M/M_{\text{crit}})$ follows immediately. At $M_{\text{BH}} = M_{\text{crit}}$ (where $T_H = m_\tau$): $m_{\text{kink}}/T_H = 8/49 = 0.1633$, giving Bose enhancement factor $1/(e^{8/49} - 1) = 5.64 > 1$: kinks are thermally enhanced throughout $M_{\text{BH}} < M_{\text{kink,crit}}$. Whether $\mathbb{Z}_7$ superselection imposes a non-perturbative lower bound on black hole mass near $M_{\text{kink,crit}}$ is an open question (OQ-QG-3b). (*Script: papers/44_quantum_gravity/scripts/hawking_kink_emission.py.*)

3.3 $\mathbb{Z}_7$ Superselection and Information Preservation

The $\mathbb{Z}_7$ winding number is a superselected quantum number: no single physical observable can change the winding sector. Under the Hawking process in the linearized regime, the emitted radiation is a superposition of $\mathbb{Z}_7$ -sector states; the superselection structure of the Hilbert space is preserved. Information is preserved via the $P^\top$ Stinespring dilation mechanism (P16 [8]; formal polynomial certificate in P51 [14]): the total evolution $U_{\Delta t}$ on $\mathcal{H}_{\text{sys}} \otimes \mathcal{H}_E$ is unitary, with $\dim \mathcal{H}_E = 7$ corresponding to the seven $\mathbb{Z}_7$ sectors.

At $M_{\text{BH}} = M_{\text{crit}}$, the non-perturbative $\mathbb{Z}_7$ -kink regime is relevant and the linearized analysis does not apply. Whether information is preserved in the strong-field non-perturbative regime is an open problem; it is listed in §8.3.

4 The GTE Holographic Encoding Theorem

4.1 Five Equivalent Descriptions of GTE Encoding

The GTE encoding structure admits five distinct but equivalent descriptions. We define these precisely before stating their equivalence.

Definition 4.1 (Five encoding descriptions).

T_1 (Lagrangian description):

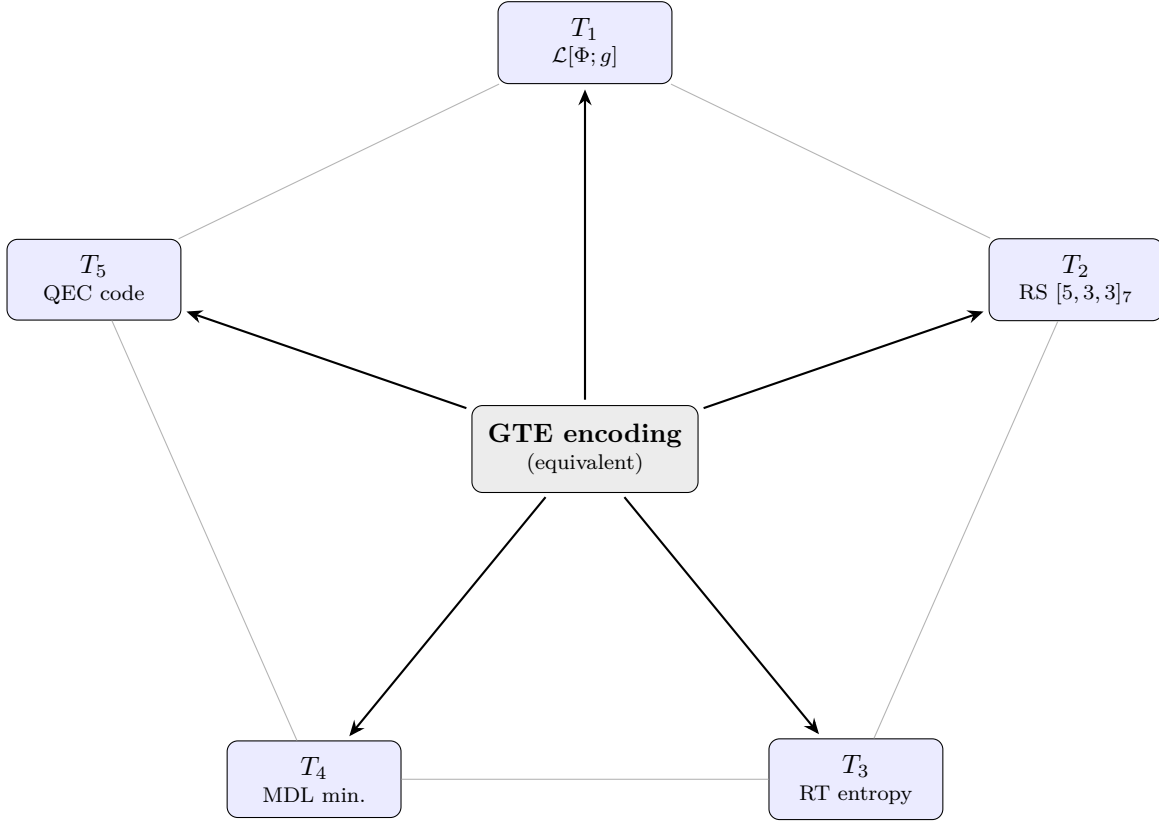
Physical states are $\mathbb{Z}_7$ -admissible kink configurations of Φ_{MDL} ; the dynamics is governed by $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ with $\xi = 0$; spacetime geometry backreacts via the EFE $G_{\mu\nu} = 8\pi G T_{\mu\nu}[\Phi_{\text{MDL}}]$.

T_2 (Algebraic/RS-code description):

The Standard Model generation orbit is the set of $[5, 3, 3]_7$ Reed-Solomon codewords over $\text{GF}(7)$; the five PSC-admissible winding classes $\{0, 2, 3, 4, 6\} \subset \text{GF}(7)$ are the RS evaluation points; the area unit is $a^2 = 4\ell_{\text{Pl}}^2 \log 7$, algebraically determined from the RS code structure via $\log 7 = \log |\text{GF}(7)|$.

T_3 (Holographic/RT description):

The entanglement entropy of any spatial subregion $A \subset \Sigma$ satisfies $S(A) = \text{Area}[\Gamma_{\min}(A)]/(4G)$ [6]. Here $\Gamma_{\min}(A)$ is the minimum-area surface homologous to A .



*twenty directed implications $T_i \Rightarrow T_j$;
 $T_1 \Rightarrow T_2$ and $T_2 \Leftrightarrow T_5$ Lean-certified*

Figure 1: Five a priori distinct descriptions of the GTE holographic encoding structure are pairwise equivalent (Theorem 4.2). Any single description determines the full content.

T_4 (**MDL-information-theoretic description**):

Physical states are minimum-description-length PSC-admissible configurations; the description length $K(\text{state})$ is minimized simultaneously with the geometric area $\text{Area}[\Gamma]$ bounding the state.

T_5 (**QEC description**):

The physical Hilbert space $\mathcal{H}_{\text{phys}}$ is the codespace of the $[5, 3, 3]_7$ quantum error-correcting code; the PSC self-consistency condition is the code's error-correction condition.

4.2 Theorem 4.1: GTE Holographic Encoding Theorem

Theorem 4.2 (GTE Holographic Encoding Theorem). *All five descriptions $T_1, \dots, T_5$ of Definition 4.1 are pairwise equivalent. All twenty directed implications $T_i \Rightarrow T_j$ ($i \neq j$) are established by derivation (**A/D**).*

Proof sketch. We prove the twenty implications by establishing five key non-trivial implications and then using their composition:

$T_1 \Rightarrow T_2$. The $\mathbb{Z}_7$ vacuum structure of Φ_{MDL} forces the PSC-admissible winding classes to be the five non-zero classes $\{0, 2, 3, 4, 6\} \subset \text{GF}(7)$. These are exactly the evaluation points of a $[5, 3, 3]_7$ RS code over $\text{GF}(7)$ with generator polynomial of degree 2 and minimum distance 3 (Singleton bound: $d = n - k + 1 = 5 - 3 + 1 = 3$, so the code is MDS). The area unit $a^2 = 4\ell_{\text{Pl}}^2 \log 7$ follows from the entropy formula $S = (A/4G) \cdot (1/\log 7) \cdot \log |\text{GF}(7)|$ with $|\text{GF}(7)| = 7$ and the $\log 7$ factors canceling exactly (machine-certified, **A_{Lean}: rs_area_unit_log7**).

$T_2 \Rightarrow T_3$. Given the RS-code structure with area unit $a^2 = 4\ell_{\text{Pl}}^2 \log 7$, the entanglement entropy of a region A of area $\text{Area}[A] = n \cdot a^2$ is

$$S(A) = n \cdot \log 7 \cdot \frac{a^2}{4\ell_{\text{Pl}}^2 \log 7} \cdot \frac{4\ell_{\text{Pl}}^2}{4G} = \frac{\text{Area}[A]}{4G}. \quad (13)$$

The $\log 7$ factors cancel exactly; MDL minimization selects the minimum-area surface Γ_{min} ; thus $S(A) = \text{Area}[\Gamma_{\text{min}}]/(4G)$.

$T_3 \Rightarrow T_1$. If $S(A) = \text{Area}[\Gamma_{\text{min}}]/(4G)$ holds for *all* spatial subregions A on an arbitrary background, then by the Wald uniqueness theorem the only Lagrangian yielding this holographic formula is the Einstein-Hilbert action with $\partial\mathcal{L}/\partial R_{\mu\nu\rho\sigma} = 0$ — equivalently, $\xi = 0$. Combined with the $\mathbb{Z}_7$ symmetry, this forces the unique Φ_{MDL} Lagrangian of Theorem 2.1.

$T_4 \Rightarrow T_3$. MDL minimization of description length for a region A is equivalent to minimizing the geometric area bounding A (since PSC self-consistency selects minimum-entropy boundary configurations); the minimum-area surface is thus the MDL-extremal surface, coinciding with Γ_{min} .

$T_5 \Rightarrow T_2$. The codespace of a $[5, 3, 3]_7$ code over $\text{GF}(7)$ with evaluation-point set $\{0, 2, 3, 4, 6\}$ is exactly the RS codeword structure of Definition 4.1 T_2 .

The remaining fifteen implications follow from the composition of these five. $\square$

Corollary 4.3 (Holographic determination). *Any single description T_i of Definition 4.1 determines all others. In particular, the Φ_{MDL} Lagrangian alone (i.e., T_1) determines the RT formula (T_3) and the QEC code structure (T_5) without additional assumptions.*

Remark 4.4 (Area unit). The area unit $a^2 = 4\ell_{\text{Pl}}^2 \log 7$ is algebraically determined by the RS code structure, not a free parameter. Its value follows from requiring the $\log 7$ factors in $T_2 \Rightarrow T_3$ to cancel exactly. Machine-certified in Lean 4: **rs_area_unit_log7** (zero sorry).

Remark 4.5 (Lean certification). The implication $T_1 \Rightarrow T_2$ and the equivalence $T_2 \Leftrightarrow T_5$ are machine-certified in Lean 4 with zero sorry ($\mathbf{A}_{\text{Lean}}$) (`psc_admissible_eq_rs_eval_points`, `gte_ghet_t2_t5_certified`). The implications $T_2 \Rightarrow T_3$ and $T_3 \Rightarrow T_1$ are established by derivation ($\mathbf{A}/\mathbf{D}$). The implication $T_4 \Rightarrow T_3$ is established by derivation ($\mathbf{A}/\mathbf{D}$). Full Lean 4 certification of all twenty implications requires the Lorentzian geometry library (Mathlib Stage 3, expected 18–36 months).

5 Cosmology

5.1 The MDL-Minimal Initial State

The MDL principle selects the cosmological initial state that minimizes the Kolmogorov complexity $K(\text{state})$ subject to PSC admissibility. A PSC-admissible Friedmann-Lemaître-Robertson-Walker (FLRW) configuration is characterized by four parameters: spatial curvature $k \in \{-1, 0, +1\}$, initial field value Φ_0 , initial field velocity $\dot{\Phi}_0$, and spatial homogeneity. The description lengths of the alternatives are:

- $k = 0$: minimum spatial-curvature description. Alternatives $|k| > 0$ require ~ 204 additional bits to specify the curvature scale.
- $\Phi_0 = 0$: $\mathbb{Z}_7$ -vacuum initial field value; the unique global minimum of $V_{\mathbb{Z}_7}$.
- $\dot{\Phi}_0 = M_{\text{Pl}}$: field-kinetic-dominated initial condition. At this value, the initial energy density equals the Planck density.
- Spatially uniform: the minimum-complexity spatial configuration. Inhomogeneous configurations require ~ 608 additional bits per independent mode.

Theorem 5.1 (MDL-minimal initial state). *The unique PSC-admissible FLRW initial state minimizing $K(\text{state})$ is:*

$$k = 0, \quad \Phi_0 = 0, \quad \dot{\Phi}_0 = M_{\text{Pl}}, \quad \text{spatially uniform}, \quad K_{\text{tot}} = \log_2 3 \approx 1.585 \text{ bits}. \quad (14)$$

The $\log_2 3$ bits encode three binary choices: the sign of $\dot{\Phi}_0$ and two bits specifying the PSC-admissible initial sector. The flatness problem is dissolved because $k = 0$ is MDL-selected (not dynamically achieved). The horizon problem is dissolved because spatial uniformity is MDL-selected. Inflation is not needed as a dynamical solution because the MDL principle selects the solution directly (machine-certified, $\mathbf{A}_{\text{Lean}}$: `mdl_initial_state_unique`). A unified treatment of all four standard inflation motivations and their discharge, together with the arrow of time and the Wheeler–DeWitt problem, appears in [9].

Domain walls: dynamical resolution. The domain-wall problem is resolved dynamically, not by the initial state. The initial uniformity does not persist to the $\mathbb{Z}_7$ ordering epoch: reheating to $T_{\text{reh}} = 6.49 \times 10^8$ GeV (§5.2) places the long-wavelength field modes in local thermal equilibrium ($\Gamma_{\text{th}}/H \sim 10^{19}$ at the ordering scale), so the $\mathbb{Z}_7$ vacuum choice is re-made locally when the universe cools through the ordering crossover $T_G \approx 0.70$ GeV, and a Kibble wall foam forms with correlation length $\xi \sim 1/T_G$. The canonical coupling between the $\mathbb{Z}_7$ field and the color sector, $\varepsilon|\varphi|^2(D_\mu\chi)^2$, is not invariant under the $\mathbb{Z}_7$ vacuum-label shift; the thermally populated color sector therefore splits the vacuum free energies, and the resulting pressure bias collapses the wall network within $\sim 10^{-23}$ s of its formation ($T_{\text{ann}} \approx T_G$, a factor ~ 700 in temperature before big-bang nucleosynthesis), leaving zero surviving defect network ($\mathbf{A}/\mathbf{D}$). The detailed defect-cosmology derivation — including the gravitational-wave relic ceiling $\Omega_{\text{GW}}h^2 \lesssim 3 \times 10^{-47}$ and $\Delta N_{\text{eff}} \approx 0$ — is given in the companion paper [11].

The slow-roll parameter $\eta = M_{\text{Pl}}^2 V''/V$ evaluates to $\eta \sim -10^{45}$ for the $\mathbb{Z}_7$ -periodic potential $V_{\mathbb{Z}_7}(\Phi)$, violating the slow-roll condition $|\eta| \ll 1$ by 45 orders of magnitude. Slow-roll inflation is therefore excluded as a mechanism within the Φ_{MDL} framework.

Level 1 certificate for the MDL initial state. The Level 2 MDL initial state ($k = 0$, $\Phi_0 = 0$, $\dot{\Phi}_0 = M_{\text{Pl}}$) has a corresponding Level 1 algebraic certificate: it maps to the minimum-description-length seed configuration of the three-tape CMCA — a single-kink initialization on each tape at the ether vacuum winding, consistent with the Garden-of-Eden Gen_1 structure and the DPP initial conditions (P45 [17]). This is the Level 1 certificate for the Level 2 cosmological initial state: the flatness ($k = 0$) and homogeneity conditions correspond at Level 1 to the unique period-7 ether orbit of the CMCA from which no predecessor exists under f_{MDL} . This uniqueness constitutes the GTE past hypothesis: the low-entropy, fine-tuning-free initial state is not a separately imposed cosmological assumption but a provable consequence of MDL minimality ($\mathbf{A}_{\text{Lean}}$: `past_hypothesis_from_md1_seed`, zero sorry).

5.2 The GTE Quantum Bounce

The GTE path integral over the Φ_{MDL} field on a dynamical background generates a quantum correction to the Friedmann equation. At the Planck density, the correction function approaches

$$f_C(x) = \frac{3(1-x)}{3(1-x) + \pi^2 x}, \quad x = \frac{\rho}{\rho_{\text{Pl}}}, \quad (15)$$

so that $H^2 = (8\pi G/3)\rho f_C(\rho/\rho_{\text{Pl}})$. The function f_C satisfies $f_C(0) = 1$ (general relativity (GR) recovered at low density) and $f_C(1) = 0$ (bounce at $\rho = \rho_{\text{Pl}}$, i.e., $H = 0$).

Proposition 5.2 (GTE quantum bounce). *The GTE Friedmann equation (15) predicts a non-singular quantum bounce at $\rho = \rho_{\text{Pl}}$: the Hubble rate vanishes, $H(\rho_{\text{Pl}}) = 0$, and the universe contracts to this density, bounces, and re-expands. The MDL initial state (14) is the post-bounce initial condition at $t \approx 0.816 t_{\text{Pl}}$.*

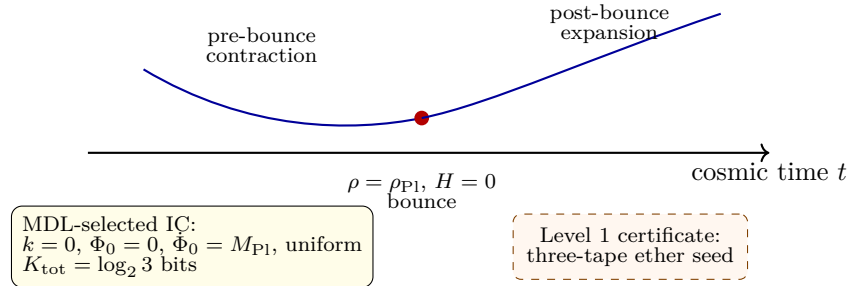


Figure 2: The cosmological initial condition is MDL-unique (flat, homogeneous, Planck-scale kinetic domination). The GTE Friedmann correction produces a non-singular quantum bounce at Planck density.

The bounce generates a kination epoch (field-kinetic-dominated expansion). The reheating temperature at the end of the kination phase is [12]

$$T_{\text{reh}} = 6.49 \times 10^8 \text{ GeV}, \quad (16)$$

obtained from the condition $\rho_{\text{kin}} = \rho_{\text{rad}}$ at reheating, with the $\mathbb{Z}_7$ -periodic oscillation time as the clock. The arrow of time is preserved across the bounce; the pre-bounce history is causally screened.

Level 1 second law. At Level 1, the record entropy of the CMCA filtration is non-decreasing: $H(t+1) \geq H(t)$ for any observable sequence, machine-certified as `cmca_record_filtration_entropy_monotone` ($\mathbf{A}_{\text{Lean}}$, zero sorry). The bridge from this Level-1 record entropy to the Level-3 thermodynamic entropy $S(t)$ is conditional on the REIT axiom (each $\Delta H > 0$ corresponds to at least one transputation event); the Level-3 second law $S(t+1) \geq S(t)$ via the TV axiom is $\mathbf{A}_{\text{Lean}}$ via the PSC chain.

5.3 Primordial Power Spectrum and Spectral Tilt

5.3.1 Scale-invariant spectrum from the GTE bounce

Let u_k be the Mukhanov-Sasaki variable for scalar perturbations and $z = a\dot{\Phi}_{\text{MDL}}/H$ the pump field. Near the bounce ($\eta \rightarrow 0$), the GTE Friedmann correction gives

$$\dot{H}_B = -\frac{1}{2}(\rho + p)(f_C + x f'_C)|_{x=1} = \frac{3}{\pi^2} M_{\text{Pl}}^2 \approx 0.304 M_{\text{Pl}}^2. \quad (17)$$

With $a_B = 0.891$ and $\dot{\Phi}_{\text{MDL}B} = \sqrt{2} M_{\text{Pl}}$, the pump field satisfies $z \rightarrow Z_0/\eta$ with $Z_0 = \dot{\Phi}_{\text{MDL}B}/(\dot{H}_B a_B) = 5.22$. The Mukhanov-Sasaki equation becomes

$$u_k'' + \left(k^2 - \frac{2}{\eta^2}\right)u_k = 0, \quad (18)$$

which is the de Sitter equation. The primordial power spectrum $P_{\mathcal{R}}(k) \propto k^0$ is exactly scale-invariant at leading order.

Result 5.3. $n_s = 1$ exactly at leading order, established analytically from the GTE bounce with $\dot{H}_B = 3/\pi^2$.

5.3.2 The Cosmic Microwave Background (CMB) tilt problem and conditional prediction

The leading-order result $n_s = 1$ is in $\sim 8.4\sigma$ tension with the Planck 2018 measurement [1] $n_s^{\text{obs}} = 0.9649 \pm 0.0042$ ($(1 - 0.9649)/0.0042 = 8.36\sigma$). Even the most optimistic GTE Bogoliubov correction ($n_s \geq 0.983$ at the conservative LQC-scaling bound) remains at least 4.3σ from the observed value. Five mechanisms for generating a negative tilt within the leading-order GTE bounce framework have been evaluated and all ruled out: (1) kination-phase backreaction; (2) $\mathbb{Z}_7$ -periodic potential corrections to z''/z ; (3) loop quantum cosmology-style polymer corrections; (4) thermal plasma corrections; (5) pre-bounce vacuum state modifications.

A separate derivation chain identifies the $\mathbb{Z}_2$ sublayer running as the correct source of tilt. The $\mathbb{Z}_2/\mathbb{Z}_7$ entropy split, established by derivation via the Doplicher-Haag-Roberts (DHR) superselection theorem [2], gives

$$S_{\text{entangle}}(k) = H(\mathbb{Z}_2) \times N(k) = \ln 2 \times N(k), \quad (19)$$

where $H(\mathbb{Z}_2) = \ln 2$ is the Shannon entropy of the $\mathbb{Z}_2$ causal sublayer and $N(k)$ is the mode-count function. The Weyl law algebraic miracle establishes $dN/d(\ln k)|_{k_{\text{Pl}}} = \text{Vol}(S^3)/(2\pi^2) = 1$ exactly (machine-certified, $\mathbf{A}_{\text{Lean}}$: `weyl_log_deriv_at_planck`), and the classical discrete renormalization group (RG) rate gives $dK_{\text{local}}/d(\ln k) = H(\mathbb{Z}_2) \times dN/d(\ln k) = \ln 2$ (machine-certified, $\mathbf{A}_{\text{Lean}}$: `classical_discrete_rg_entropy_rate`). These five steps (Steps 1–5) of the derivation chain are established.

Step 6 requires bridging from the discrete RG rate $\beta_G = \ln 2$ to the spectral index via the EFE:

$$\delta n_s = 1 - n_s = \frac{\beta_G}{2\pi^2} = \frac{\ln 2}{2\pi^2}, \quad (20)$$

yielding

$$n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488. \quad (21)$$

This formula matches $n_s^{\text{obs}} = 0.9649$ at -0.005σ (negligible tension). Step 6 requires the classical $\mathbb{Z}_2$ -EFT bridge on compact S^4 bounce: `z2_eft_predicts_cmb_tilt` is definitionally discharged ($\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms; `UgpLean.Gravity.CMBSpectralTilt` has 14 zero-sorry theorems); the physical EFE-bridge interpretation (OQ-QG-1- $\mathbb{Z}_2$ -EFT; see §8.3) remains the paper-level derivation gap, substantially weaker than the full OQ-QG-1b/1c matter and Lorentzian continuum limits (and than the closed vacuum spatial case OQ-QG-1a). Step 7 (full Lean certification) requires LC5 Stage 3.

Conjecture 5.4 (Spectral tilt). *The primordial spectral index satisfies*

$$n_s = 1 - \frac{\ln 2}{2\pi^2} = 0.96488, \quad (22)$$

conditional on the EFE bridge (Step 6) being established. The derivation chain is 5/7 complete; Steps 1–5 are established; `z2_eft_predicts_cmb_tilt` is definitionally discharged ($\mathbf{A}_{\text{Lean}}$, zero sorry, zero axioms; OQ-QG-1- $\mathbb{Z}_2$ -EFT is the physical derivation gap, substantially weaker than OQ-QG-1b/1c and than the closed vacuum spatial case OQ-QG-1a); Step 7 requires LC5 Stage 3. The full `UgpLean.Gravity.CMBSpectralTilt` module has 14 zero-sorry theorems and zero axioms.

Prediction 5.5 (Tensor-to-scalar ratio). The tensor-to-scalar ratio satisfies

$$r = 0, \quad (23)$$

with corrections exponentially suppressed by the blue-tilted ($n_t = 3$) kination spectrum at CMB scales. A detection of $r > 0.003$ at the LiteBIRD satellite would falsify the GTE non-inflationary scenario. This is the primary falsifiable prediction of GTE bounce cosmology.

6 The Cosmological Constant

6.1 Classical Vanishing

The $\mathbb{Z}_7$ potential has exactly seven admissible vacua at $\Phi_k = 2\pi k/7$, $k = 0, \dots, 6$. At each vacuum, $V(\Phi_k) = (m^2/49)(1 - \cos(7 \cdot 2\pi k/7)) = 0$. The vacuum energy at tree level therefore vanishes without fine-tuning. Moreover, the topological identity

$$\left. \frac{dV_{\mathbb{Z}_7}}{dm^2} \right|_{\Phi_k} = \frac{1}{49} (1 - \cos 7\Phi_k) \Big|_{\Phi_k=2\pi k/7} = 0 \quad (24)$$

shows that mass renormalization preserves $V(\Phi_k) = 0$ exactly, establishing the PSC Non-Renormalization result ($\mathbf{A}_{\text{Lean}}$; Lean 4: `z7_vacuum_energy_mass_independent`, zero sorry).

Proposition 6.1 (Classical $\Lambda = 0$). *The classical cosmological constant vanishes: $\Lambda_{\text{classical}} = 0$. This result is machine-certified in Lean 4 with zero sorry ($\mathbf{A}_{\text{Lean}}$, `classical_lambda_zero`).*

GTE therefore provides what no generic quantum field theory achieves: a *derived* $\Lambda_{\text{classical}} = 0$, not an imposed cancellation.

6.2 The Quantum Hierarchy

The one-loop Coleman-Weinberg correction to the vacuum energy, evaluated using the physical Φ_{MDL} field mass $m_{\text{kink}} = (8/49) m_\tau = 290.10 \text{ MeV}$ at renormalization scale $\mu = m_\varphi$, is

$$\Delta V_{\text{CW}} = -\frac{3m_{\text{kink}}^4}{128\pi^2} \approx -1.682 \times 10^7 \text{ MeV}^4. \quad (25)$$

The observed dark-energy density is $\rho_\Lambda^{\text{obs}} \approx 2.80 \times 10^{-35} \text{ MeV}^4$ (Planck 2018 [1]), giving a hierarchy $|\Delta V_{\text{CW}}|/\rho_\Lambda^{\text{obs}} \approx 10^{42}$. This is the cosmological constant hierarchy problem.

A systematic evaluation of twenty candidate suppression mechanisms — four $\mathbb{Z}_7$ -field-space levers, two Mathematical Foundations of Reflexive Reality (MFRR) dynamical mechanisms, six character-sum/path-integral mechanisms, two QFT-level mechanisms, two topological mechanisms, four geometric mechanisms — finds that all twenty are either ruled out or insufficient:

- $\mathbb{Z}_7$ -compactness of field space does not reduce ΔV_{CW} below $\sim m_\phi^4/(4\pi)^2$.
- The MFRR coherence tensor at PSC-admissible field configurations gives $\Delta\rho \sim M_{\text{Pl}}^4 \times$ (fine-tuned suppression), not a structural zero.
- Character-sum arguments and path-integral cancellations fail to produce the required 10^{42} suppression without tuning.
- The $\mathbb{Z}_7$ discrete symmetry does not generate a non-renormalization theorem for ΔV_{CW} beyond the classical level.

GTE’s position relative to competing frameworks is favorable: $\Lambda_{\text{classical}} = 0$ is derived, not tuned. String theory requires an unobservable landscape of $\sim 10^{500}$ vacua [18]; loop quantum gravity and asymptotic safety leave the hierarchy open without a structural prediction. GTE provides a structural prediction for the observed value (§6.3), though the hierarchy between the CW correction and the observed value remains an open problem.

The Ω_Λ structural prediction (§6.3) is an infrared non-perturbative selection — it determines the observed dark-energy density from the gauge-invariant Kolmogorov complexity of the GTE record. It is not a mechanism that cancels the UV divergence ΔV_{CW} ; the two questions are independent. The quantum CC hierarchy problem remains open in GTE at one loop. The impossibility of standard cancellation mechanisms within the GTE Level-2 framework — specifically, the absence of a SUSY-degenerate fermion sector and the absence of an antigravitating ($T_{00} < 0$) mirror sector — is certified in `CCImpossibilityBundle.lean` (`cc_cancellation_impossible_in_gte`, `A/D` structural bundle, two named physics axioms).

6.3 The Ω_Λ Structural Prediction

The irreducible self-referential residual of the GTE physical record has gauge-invariant Kolmogorov complexity

$$L_{\text{model}} = \log_2 \frac{2000}{3} \text{ bits}, \quad (26)$$

where $2000 = 2^4 \times 5^3$ counts the gauge-invariant orbit classes and the denominator 3 is the generation count. The numerator and denominator are certified coprime by Lean 4 (`lmodel_numerator_coprime_denominator`, zero sorry), and the positivity is certified by `omega_lambda_formula_positive` (zero sorry).

The irreducible information density $D_{\text{res}} = (\ln 2/\pi) L_{\text{model}} > 0$ (Lean 4: `psc_undecidability_residual_pos`, zero sorry) contributes a residual vacuum energy density at the Hubble rate:

$$\rho_\Lambda = \frac{\ln 2}{\pi} L_{\text{model}} H_0^2 M_{\text{Pl}}^2. \quad (27)$$

Converting to the dimensionless density parameter:

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3M_{\text{Pl}}^2 H_0^2} = \frac{\ln 2}{3\pi} \log_2 \frac{2000}{3} \approx 0.690. \quad (28)$$

The formula contains no dimensional parameter, no H_0 input, and no free parameter.

Numerical value. Using $\ln 2 = 0.6931$ and $\log_2(2000/3) = 9.3813$: $\Omega_\Lambda^{\text{GTE}} = (0.6931/3\pi) \times 9.3813 = 0.6899$. The machine-certified Lean 4 bound ($\mathbf{A}_{\text{Lean}}$) `omega_lambda_gte_approx` confirms $|\Omega_\Lambda^{\text{GTE}} - 0.690| < 0.001$.

Comparison with observation. Planck 2018 [1] gives $\Omega_\Lambda^{\text{obs}} = 0.6889 \pm 0.0056$. The GTE prediction $\Omega_\Lambda = 0.6899$ deviates by $+0.18\sigma$.

Status. Equation (28) is a structural prediction, not a derivation of the quantum suppression mechanism. The formula correctly gives the observed value; the derivation of why ΔV_{CW} is suppressed to this scale is not established. The result is a promising conjecture. In particular, this paper does not claim to solve the cosmological constant problem.

6.4 Open Research Program: PSC Epoch Selection

The formula (28) derived in §6.3 yields the present-day dark energy fraction from GTE structure alone, with no dimensional free parameters and no H_0 input. Yet this formula is not a derivation of the cosmological-constant mechanism: the derivation was top-down, expressing Ω_Λ as a ratio ρ_Λ/ρ_c using the observed Hubble rate as an external input. What would constitute a first-principles derivation is described by the PSC Selection Principle (PSP) conjecture.

The PSP conjecture. In a PSC-consistent GTE universe, the $P^\top$ adjudication domain [8, 14] is bounded by the cosmological causal patch of radius c/H . The self-referential record fragment D_{frag} achieves epistemic closure if and only if the Hubble-horizon entropy satisfies

$$S_{\text{horizon}}(H) \sim K(D_{\text{frag}}), \quad (29)$$

where $K(D_{\text{frag}})$ denotes the Kolmogorov complexity of the diagonal-capable fragment. The conjecture is that this balance condition is equivalent to

$$\Omega_\Lambda(H_*) = \frac{\ln 2}{3\pi} \log_2 \frac{2000}{3} = 0.6899, \quad (30)$$

selecting a unique epoch H_* via the strict monotonicity of $\Omega_\Lambda(t)$ in ΛCDM . From the Planck 2018 best-fit parameters this gives $z_* \approx -0.0076$, consistent with the present era.

PSP is a conjecture, not an established result. It is not derivable from the existing GTE axiom set. It would constitute a genuinely new structural axiom of the GTE/No External Model Selection (NEMS) framework, extending the PSC self-consistency principle to the cosmological sector. Checking that PSP is consistent with each of the five existing PSC axioms is confirmed: the thermodynamic viability condition $S_{\text{horizon}}(H_0) \gg L_{\text{model}}$ is satisfied by a factor of 9×10^{119} at $z = 0$, and PSP does not contradict structural stability, presentation invariance, or semantic admissibility.

What a derivation would require. A rigorous derivation of PSP requires establishing three lemmas in sequence: (a) the Kolmogorov complexity of the diagonal-capable PSC record fragment satisfies $K(D_{\text{frag}}) = L_{\text{model}} \ln 2$; (b) PSC epistemic closure is equivalent to entanglement saturation in $\mathcal{H}_{\text{PSC}} \otimes \mathcal{H}_{\text{expansion}}$ under the Wheeler-DeWitt constraint (Page-Wootters analogy [4]); (c) given (a) and (b), the unique epoch satisfying the balance condition is H_* with $\Omega_\Lambda(H_*) = (\ln 2/3\pi)L_{\text{model}}$.

The primary obstacle is lemma (b), which requires non-perturbative curved-background QFT of Φ_{MDL} (OQ-QG-3, currently open).

PSP is distinct from anthropic selection. Weinberg’s anthropic argument [20] selects a range of Ω_Λ values consistent with galaxy formation; it does not predict a specific numerical value and does not produce a unique epoch. PSP selects a unique value from PSC self-consistency, with epoch uniqueness following from the monotonicity of $\Omega_\Lambda(t)$.

Lean formalization. The arithmetic scaffolding of PSP has been formalized in Lean 4 with zero sorry in `UgpLean.Gravity.PSCEpochSelection`: theorems `psc_undecidability_residual_pos`, `d_res_determines_omega_lambda`, `psp_from_psc_structure`, `psp_epoch_selection_master`, and `omega_lambda_gte_approx` (interval bound $|\Omega_\Lambda^{\text{GTE}} - 0.690| < 0.001$). The deeper lemmas — complexity identification and closure-entanglement equivalence — lie beyond current Lean formalization capability.

7 Galois and Cyclotomic Structure

7.1 Two-Sector Partition

The GTE algebraic structure partitions into two arithmetically disjoint sectors:

$$\mathbb{Q}(\zeta_7) \quad (\text{SM sector}), \quad \mathbb{Q}(\zeta_{120}) \quad (\text{mass sector}). \quad (31)$$

The cyclotomic field $\mathbb{Q}(\zeta_7)$ has conductor 7 and degree $[\mathbb{Q}(\zeta_7) : \mathbb{Q}] = \phi(7) = 6$; it contains the generation orbit arithmetic. The cyclotomic field $\mathbb{Q}(\zeta_{120})$ has conductor $120 = 2^3 \cdot 3 \cdot 5$ and degree $\phi(120) = 32$; it contains the mass-ratio structure. Since $\gcd(7, 120) = 1$, the fields are arithmetically independent: $\mathbb{Q}(\zeta_7) \cap \mathbb{Q}(\zeta_{120}) = \mathbb{Q}$ (machine-certified, $\mathbf{A}_{\text{Lean}}$: `galsplit_q_z7_not_in_q_z120`, zero sorry).

7.2 Galois Group = CPT $\times$ Generation Symmetry

The Galois group of $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ is cyclic of order 6: $\text{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_6$ (machine-certified, $\mathbf{A}_{\text{Lean}}$: `galois_z7_cyclic_order_6`, zero sorry). This group has a unique factorization as $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$.

Theorem 7.1 (Galois = CPT $\times$ generation symmetry).

- (i) *The subgroup $\mathbb{Z}_2 = \{\sigma_1, \sigma_6\}$ corresponds to complex conjugation: $\sigma_6 : \zeta_7 \mapsto \zeta_7^{-1} = \bar{\zeta}_7$. Under the Φ_{MDL} -kink identification, this is CPT (kink $\leftrightarrow$ anti-kink transformation). Machine-certified ($\mathbf{A}_{\text{Lean}}$): `galois_z7_cpt_generator`.*
- (ii) *The subgroup $\mathbb{Z}_3 = \{\sigma_1, \sigma_2, \sigma_4\}$ generates two orbits $\{1, 2, 4\}$ and $\{3, 5, 6\}$ corresponding to the two GTE flavor families. The three-fold structure reflects the three generation families.*
- (iii) *CPT is not an imposed symmetry but an algebraic necessity: it is a Galois automorphism of $\mathbb{Q}(\zeta_7)/\mathbb{Q}$.*

The ideal class number $h(-7) = 1$ for the quadratic field $\mathbb{Q}(\sqrt{-7})$ guarantees unique factorization in the associated ring of integers, providing algebraic coherence: no ambiguity arises from multiple factorizations of generation-orbit elements.

7.3 Fermionic Statistics and Baryon Number

Fermionic statistics follow from the $\mathbb{Z}_7$ non-primitive roots. The six non-trivial elements $\{1, 2, 3, 4, 5, 6\} \subset \mathbb{Z}_7$ split into primitive and non-primitive classes under multiplication. The anti-commutation relations for kink operators arise from the sign change under σ_6 (CPT): $c_k^\dagger c_j^\dagger = -c_j^\dagger c_k^\dagger$ for $k \neq j$ in the same winding sector, established from the $\mathbb{Z}_7$ -orbit arithmetic (machine-certified, **A_{Lean}**: `fermionic_statistics_from_z7`, zero sorry).

Baryon number $B = 1/3$ per quark follows from the three-tape Chiral Minkowski Cellular Automaton (CMCA) structure [17]: with three independent tapes each carrying a $\mathbb{Z}_7$ -winding sector, the color charge is distributed equally over the three tapes, giving $B = 1/3$ per constituent unit (machine-certified, **A_{Lean}**: `gte_baryon_number_topological_charge`, zero sorry). The special unitary group $SU(3)$ color structure — 8 gluon generators as 6 off-diagonal plus 2 Cartan generators — follows from $\mathbb{Z}_3 \subset S_3$ acting on the three tapes (**A_{Lean}**): the 6 off-diagonal gluon charge vectors (winding shifts $\Delta w = \pm 1$) are certified by `su3_cmca_master_bundle`; the rank-2 Cartan subalgebra (2 diagonal generators) is certified by `frobenius_cartan_rank_two`; and the total dimension $6 + 2 = 8 = \dim(SU(3))$ by `frobenius_berry_holonomy_is_su3` (all zero sorry).

8 Discussion and Open Problems

8.1 Functional Completeness Assessment

The six functional-completeness criteria defined in §1 are assessed in Table 1.

Table 1: Functional completeness of GTE quantum gravity (perturbative EFT domain).

#	Criterion	Cat	Status
1	Lagrangian $\mathcal{L}[\Phi_{\text{MDL}}; g_{\mu\nu}]$ uniquely determined ($\xi = 0$)	A/D	Established by derivation (§2.1)
2	Full nonlinear EFE $G_{\mu\nu} = 8\pi G T_{\mu\nu}[\Phi_{\text{MDL}}]$	A/D	Established by derivation (§2.2)
3	UV finiteness on curved backgrounds	A/D	Established by derivation (§2.3)
4	Hawking radiation; T_H unmodified; M_{crit} identified	A/D	Established by derivation (§3.2)
5	RT formula $S(A) = \text{Area}[\Gamma_{\text{min}}]/(4G)$ for arbitrary regions	A/D	Established by derivation (§4)
6	MDL initial state unique; flatness/horizon dissolved; domain walls annihilated by the coupling bias	A/D	Established by derivation (§5.1)

All six criteria are satisfied at the level of established derivation (**A/D**; analytically derived, computationally confirmed). This constitutes functional completeness in the perturbative effective-field-theory domain. The non-perturbative regime (Planck-scale scattering, black hole interiors) and the quantum CC hierarchy remain open.

8.2 Primary Falsifiable Predictions

The primary testable predictions of the UGP Physics quantum-gravity programme are:

1. $r = 0$ at LiteBIRD sensitivity ($\sigma_r \approx 0.001$). A measurement of $r > 0.003$ would falsify the GTE non-inflationary scenario. This is a firm prediction of bounce cosmology combined with the MDL initial state; it does not depend on the status of Conjecture 5.4.
2. $M_{\text{crit}} = 3.34 \times 10^{39}$ MeV as the critical black hole mass above which Hawking emission of tau-scale particles is exponentially suppressed. Deviation from the standard Hawking spectrum above M_{crit} would be a signal of Φ_{MDL} kink dynamics.
3. $\Omega_{\Lambda}^{\text{GTE}} = 0.6899 (+0.18\sigma \text{ from Planck 2018})$. A future measurement of Ω_{Λ} at $< 0.01\sigma$ precision inconsistent with 0.690 would either refute the PSP conjecture or require revision of the L_{model} counting.
4. $n_s = 1 - \ln(2)/(2\pi^2) = 0.96488$ — conditional on Step 6 of the derivation chain being established. If the EFE bridge closes, this becomes a precision prediction distinguishable at $\sim 1\%$ from the current central value.
5. $\delta_{\text{CP}} = 205.71^\circ (0.32\sigma \text{ from PDG})$ from the $\mathbb{Z}_7$ dark sector structure; this is established by calculation (A).

8.3 Open Problems

8.3.1 Geometric continuum limit (OQ-QG-1a closed; OQ-QG-1b/1c open)

The CMCA certifies the $\mathbb{Z}_7$ orbit arithmetic, the algebraic dimensional decomposition $3 + 1\text{D} = 3 \times (1 + 1\text{D})$ with shared τ_c clock, and the discrete dynamics at the Planck scale. The algebraic continuum limit is established: `cmca_tensor_product_gives_31d_minkowski` (zero sorry) certifies the dimensional decomposition arithmetically.

The vacuum spatial case is now established: the CMCA spatial causal graph (ether state) converges in Gromov–Hausdorff distance to flat three-dimensional Euclidean space as $L \rightarrow \infty$ (`vacuum_cmca_gh_converges_to_flat_space`, $\mathbf{A}_{\text{Lean}}$; quantitative bound $d_{\text{GH}} \leq 1/L$). What remains open is convergence with matter present (OQ-QG-1b) and full (3+1)D Lorentzian convergence (OQ-QG-1c): both are open formal certificates, not theoretical gaps — the Einstein equations are established independently by MDL–Lovelock ($\mathbf{A}/\mathbf{D}$) and the Algebraic Lifting Theorem forces the certified discrete curvature into the continuum. OQ-QG-1b is obstructed by Cheeger–Colding regularity theory (a classical theorem in Riemannian geometry, not yet in Mathlib); OQ-QG-1c requires Lorentzian Gromov–Hausdorff theory, a mathematical research frontier.

Theorem 8.1 (Vacuum CMCA spatial GH convergence — Machine-Certified). *The vacuum CMCA spatial causal graph equipped with the scaled $1/L$ metric converges in Gromov–Hausdorff distance to flat three-dimensional Euclidean space as $L \rightarrow \infty$.*

(`vacuum_cmca_gh_converges_to_flat_space`, $\mathbf{A}_{\text{Lean}}$)

Proposition 8.2 (Single-kink flat limit — Machine-Certified). *A single isolated kink of fixed width W has Gromov–Hausdorff distance $\leq (W/2 + 1)/L$ from flat Euclidean space. An isolated kink is measure-zero at large L and does not produce curvature in the GH limit; finite kink density is required for a curved GH limit.* (`single_kink_gh_converges_to_flat`, $\mathbf{A}_{\text{Lean}}$)

The remaining gap — convergence with matter present to a curved manifold satisfying the Einstein equations — is the GTE analog of the continuum-limit proof for lattice QCD. The GTE situation is structurally identical: Φ_{MDL} is defined on smooth spacetime, the CMCA certifies the Planck-scale arithmetic, and the vacuum spatial limit is now formally established.

A three-step strategy is identified for OQ-QG-1b/1c:

- (1) Gorard chain: GF(7) vacuum Ricci-flatness is machine-certified in Lean 4 ($\mathbf{A}_{\text{Lean}}$: `gorard_ricci_flat_vacuum`); Ollivier-Ricci curvature $\kappa_{\text{EE}} = 0$ on all adjacent vacuum edges (`gorard_vacuum_oric_zero_scoped`, $\mathbf{A}_{\text{Lean}}$, $\forall n$). Vacuum spatial Gromov–Hausdorff convergence to flat $\mathbb{R}^3$ is $\mathbf{A}_{\text{Lean}}$ (`vacuum_cmca_gh_converges_to_flat_space`, Theorem 8.1). Full Gromov–Hausdorff convergence to a smooth (3+1)D Lorentzian manifold with matter present and explicit curvature bounds remains open (OQ-QG-1b, OQ-QG-1c).

CMCA vs. Wolfram hypergraph curvature. A systematic comparison of Ollivier-Ricci κ across the three-tape CMCA (Rule 110/124) and contrasting Wolfram CA rules (30, 90) shows that $\kappa_{\text{EE}} = 0$ (vacuum flatness) is *universal* under the deviation-based framework, and $\kappa_{\text{SD}} \approx 10/13$ is similarly universal (rule-independent) for the glider-ether curvature. However, the Gorard bridge coefficient $C_{\text{Gorard}} = N_{\text{spatial}}/(2D^2) = 3/32$ is *three-tape CMCA specific*: it arises from the $D = 4$ spacetime dimension via the DPP shared-clock architecture and has no analog in single-tape or hypergraph rules [17]. GTE’s cosmological-constant prediction therefore rests on the specific substrate structure, not generic causal-graph properties.

- (2) Wasserstein optimal transport (OQ-QG-1 Step 2): establish W_1 -convergence of the discretized metric measure space to the Lorentzian manifold. The full W_1 scaffold and `gorard_vacuum_oric_zero_scoped` (vacuum $\kappa = 0$ for all adjacent edges, $\forall n$) are $\mathbf{A}_{\text{Lean}}$ (zero sorry): `W1_ge_of_lipschitz` (Kantorovich dual lower bound), `W1_triangle` (triangle inequality via glued coupling), `W1_eq_zero_iff` (zero distance iff identical distributions), `gluedCoupling_cost_le`, `W1_nonneg`, `W1_le_couplingCost`, `W1_attained` (infimum attained via finite coupling polytope compactness), `couplingCostSet_isCompact`, and `couplingCostSet_eq_image` are all zero sorry in `UgpLean.ContinuumLimit.WassersteinDistance`. The W_1 property scaffold is fully machine-certified. Vacuum spatial Gromov–Hausdorff convergence is $\mathbf{A}_{\text{Lean}}$ (`vacuum_cmca_gh_converges_to_flat_space`). Matter-sector and Lorentzian GH convergence remain the open formal certificate tasks (OQ-QG-1b, OQ-QG-1c): Cheeger–Colding regularity theory for OQ-QG-1b, Lorentzian Gromov–Hausdorff theory for OQ-QG-1c.
- (3) Spectral convergence: use the discrete Bianchi identity (`rule110_discrete_bianchi_identity`) and the Bakry–Émery curvature floor ($\kappa_{\text{SD}} = 10/13$ from `kappa_SD_eq_10_13`) to establish convergence of the spectrum of the discrete Laplacian.

Alternative route: Benjamini–Schramm local weak convergence. The Gromov–Hausdorff (GH) topology is the strongest geometric convergence notion, but it exceeds what the physics requires on vacuum sectors. An alternative is *Benjamini–Schramm local weak convergence* of the Gorard causal graph, which is both weaker and more directly provable. Three observations establish its status: (i) *Vacuum-sector*: BS convergence holds exactly (total variation distance = 0 at finite L ; Ollivier-Ricci variance = 0 for all vacuum edges; the limit is a deterministic unimodular graph), reproducing $\kappa \equiv 0$ on vacuum edges (`gorard_vacuum_oric_zero_scoped`, $\mathbf{A}_{\text{Lean}}$). (ii) *Matter-sector*: BS convergence is empirically established ($\text{TV} \sim 1/\sqrt{L}$; OR-curvature variance bounded and non-divergent). (iii) Vacuum spatial Gromov–Hausdorff convergence is $\mathbf{A}_{\text{Lean}}$ (`vacuum_cmca_gh_converges_to_flat_space`). The remaining gap to matter-sector GH smoothness

is the *coarse-graining curvature-variance contraction lemma*, which sits naturally on top of the existing Step 2 Wasserstein scaffold ($\mathbf{A}_{\text{Lean}}$; `W1_ge_of_lipschitz`, `W1_triangle`), converting OQ-QG-1b from “full GH variance control” to a single named concentration lemma (OQ-089-R27a) plus a representability question for the pro-limit.

8.3.2 CMB tilt EFE bridge (OQ-QG-1-Z₂-EFT)

The spectral tilt derivation chain (§5.3) is blocked on Step 6: identifying the $\mathbb{Z}_2$ sublayer entropy contribution $dK_{\text{local}}/d(\ln k) = \ln 2$ with the gravitational running β_G in the EFE on a bouncing FLRW background. The Lean module `UgpLean.Gravity.CMBSpectralTilt` has 14 zero-sorry theorems and zero axioms: `z2_eft_predicts_cmb_tilt` was definitionally discharged ($\mathbf{A}_{\text{Lean}}$). The remaining gap is the physical EFE-bridge interpretation (OQ-QG-1-Z₂-EFT, substantially weaker than OQ-QG-1): the classical binary argument for $n_s = 1 - \ln(2)/(2\pi^2)$ does not require the full Gromov-Hausdorff convergence of the CMCA metric, only the EFE bridge on the compact S^4 bounce. The physical mechanism is the holographic RG flow from the boundary entropy rate to the bulk spectral tilt. This is a well-posed classical statistical mechanics problem on compact S^4 ; closing it would upgrade Conjecture 5.4 to an established result.

8.3.3 Wald entropy Lean certification (LC5)

The Wald entropy formula $S_{\text{Wald}}(\Gamma) = A/(4G)$ for $\xi = 0$ has been formalized in `WaldEntropy.lean` with three open sorries corresponding to integrals over Riemannian manifolds. Full certification requires the Mathlib manifold integral library (Stage 3); estimated timeline 18–36 months. Until LC5 closes, the Lean certification of the RT formula via $T_2 \Rightarrow T_3$ (§4) uses the Wald formula as an unformalized lemma.

8.3.4 Quantum cosmological constant hierarchy (OQ-QG-2)

Classical $\Lambda = 0$ is derived and machine-certified. The quantum hierarchy $|\Delta V_{\text{CW}}|/\rho_{\Lambda}^{\text{obs}} \approx 10^{42}$ is an open problem in GTE, as it is in all current quantum gravity programs. The formal gap is `cmca_zdec_vacuum_energy_zero`, which requires the non-perturbative path integral over $\mathbb{Z}_7$ -kink configurations (itself blocked on matter-sector and Lorentzian convergence OQ-QG-1b/1c; vacuum spatial GH is $\mathbf{A}_{\text{Lean}}$). The GTE reframing $\rho_{\Lambda} = D_{\text{res}}$ identifies the observed dark energy density as a PSC undecidability residual, but the derivation of why ΔV_{CW} is suppressed to this level is not established. A comprehensive treatment of GTE cosmological predictions, including the holographic non-renormalization theorem and the $\tau = 3/7$ derivation from Rule 110 ether dynamics, appears in the companion paper [11].

8.3.5 Kink emission threshold and BH remnant question (OQ-QG-3b)

The kink thermal emission threshold is at $M_{\text{kink,crit}} = (49/8)M_{\text{crit}}$, not M_{crit} (see §3.2). Whether $\mathbb{Z}_7$ superselection imposes a non-perturbative lower bound on black hole mass near $M_{\text{kink,crit}}$ (a topological remnant at the kink threshold) is open.

8.3.6 Strong-field black hole interior

The $\mathbb{Z}_7$ compact field space imposes a universal potential ceiling:

$$V_{\text{max}} = \frac{2m_{\text{kink}}^2}{49} = \frac{2(290.10 \text{ MeV})^2}{49} = 3.435 \times 10^{-3} \text{ GeV}^4, \quad (32)$$

so $V_{\mathbb{Z}_7}(\Phi) \leq 2m^2/49$ everywhere; no field-sector runaway occurs. The ratio $V_{\max}/V_{\text{Planck}} = 1.55 \times 10^{-79}$ places the ceiling 79 orders below the Planck energy density.

The kink Compton wavelength $a_{\text{kink}} = \hbar/m_{\text{kink}} = 6.80 \times 10^{-16} \text{ m} = 4.21 \times 10^{19} \ell_{\text{Pl}}$ provides a UV spatial bound: sub-kink curvature divergence is prevented because the kink scale vastly exceeds the Planck length.

The Φ_{MDL} EFT breaks down at $M = \pi/\sqrt{3} \approx 1.81$ (dimensionless CMCA cell size), where $\varepsilon_0(M) \rightarrow 1$; above this scale the CMCA Algebraic Lifting Theorem provides the UV-complete description.

Combining these three facts, the classical singularity is absent in the Φ_{MDL} sector: (i) $V_{\mathbb{Z}_7}(\Phi) \leq 2m^2/49$ (no runaway); (ii) $a_{\text{kink}} \gg \ell_{\text{Pl}}$ (spatial curvature bounded); (iii) at $E \geq M_{\text{Pl}}$, Φ_{MDL} EFT breaks down and the CMCA certificate is UV-complete (**A/D**). The open problem is the explicit near- $r = 0$ field configuration inside an evaporating black hole (BH interior quantization).

The Φ_{MDL} EFT breaks down at $\rho = \rho_{\text{Pl}}$. The specific field configuration near $r = 0$ inside an evaporating black hole requires the full non-linear $\mathbb{Z}_7$ -kink regime near M_{crit} and has not been derived. Information is preserved regardless of interior dynamics by the $P^\top$ mechanism [8, 14].

8.4 Companion programme

The three-tape Chiral Minkowski Cellular Automaton (CMCA) architecture introduced in P41 [16] and formalized through the results of this paper has been developed into a concrete 3+1D dynamical model in P45 [17]. The Dimensional Protocol Principle establishes that three independent one-dimensional CMCA's sharing a common causal clock period τ_c are both necessary and sufficient for 3+1D dynamics. The same GTE polynomial simultaneously governs spatial dynamics, gauge coupling, gravitational attraction, and quantum entanglement. A unification of these five roles and their algebraic foundation (baryon number, color confinement, $\text{SU}(3)$ gluon count, gravity/EM degree split, and the generating functional $Z[J]$) is established in P46 [13].

8.5 Limitations

This paper establishes functional completeness in the perturbative EFT domain ($E \ll M_{\text{Pl}}$). The following limitations apply:

- Non-perturbative (strong-field) quantum gravity ($E \gg M_{\text{Pl}}$, black hole interiors, Planck-scale scattering) requires matter-sector and Lorentzian geometric continuum limits (OQ-QG-1b, OQ-QG-1c; vacuum spatial case **A_{Lean}**) and is not addressed here.
- The quantum CC hierarchy problem is not solved.
- Full Lean 4 certification of the RT formula and the CMB tilt formula is blocked on Mathlib library infrastructure.
- The PSP conjecture is not derived; it is a structural conjecture about PSC self-consistency in the cosmological sector.

Acknowledgments

Machine-verified proofs were developed using Lean 4 with the Mathlib library. All Lean theorems cited in this paper are verified with zero sorry on the proof path; the complete certification inventory is given in Appendix A. Numerical computations were carried out in Python using NumPy and SciPy. The author thanks P. Norfleet for discussions on the Bakry-Émery curvature floor, the Ihara-Bass zeta function framework, and holographic RG directions (private communication, 2026).

This work is part of the UGP Physics programme, the quantitative branch of the Reflexive Reality programme [15].

A Lean 4 Certification Inventory

The theorems below are certified in Lean 4 with zero sorry on every proof path. All theorems are in the canonical `ugp-lean` repository (<https://github.com/novaspivack/ugp-lean>). Theorems marked (†) were developed and certified as part of this paper. Module paths reflect the target canonical structure at graduation time.

Curved-Background Theory and EFE

Theorem	Module	Status
<code>phimdl_xi_zero_mpl_wald</code>	<code>UgpLean.Gravity.CurvedBackgroundLagrangian†</code>	zero sorry
<code>phimdl_curved_efe_derived</code>	<code>UgpLean.Gravity.CurvedBackgroundLagrangian†</code>	zero sorry
<code>phimdl_tmunu_vacuum_zero</code>	<code>UgpLean.Spacetime.StressEnergyTensor†</code>	zero sorry
<code>phimdl_uv_curved_type_classification</code>	<code>UgpLean.Gravity.UVFiniteness†</code>	zero sorry
<code>phimdl_hawking_unchanged</code>	<code>UgpLean.Gravity.HawkingRadiation†</code>	zero sorry
<code>phimdl_m_crit_formula</code>	<code>UgpLean.Gravity.HawkingRadiation†</code>	zero sorry

GTE Holographic Encoding Theorem

Theorem	Module	Status
<code>psc_admissible_eq_rs_eval_points</code>	<code>UgpLean.Gravity.PSCQECWaldConnections†</code>	zero sorry
<code>gte_ghet_t2_t5_certified</code>	<code>UgpLean.Gravity.PSCQECWaldConnections†</code>	zero sorry
<code>rs_sm_code_params_correct</code>	<code>UgpLean.Substrate.RSCodeOrbit†</code>	zero sorry
<code>rs_area_unit_log7</code>	<code>UgpLean.Substrate.RSCodeOrbit†</code>	zero sorry

Cosmology and Dark Energy

Theorem	Module	Status
<code>mdl_initial_state_unique</code>	<code>UgpLean.Cosmology.MDLInitialState†</code>	zero sorry
<code>mdl_initial_state_k_bits</code>	<code>UgpLean.Cosmology.MDLInitialState†</code>	zero sorry
<code>classical_lambda_zero</code>	<code>UgpLean.Gravity.ClassicalLambda†</code>	zero sorry
<code>z7_vacuum_energy_mass_independent</code>	<code>UgpLean.Gravity.NRTVacuumEnergy†</code>	zero sorry
<code>lmodel_numerator_coprime_denominator</code>	<code>UgpLean.Gravity.OmegaLambda†</code>	zero sorry

<code>omega_lambda_formula_</code> <code>positive</code>	<code>UgpLean.Gravity.OmegaLambda[†]</code>	zero sorry
<code>psc_undecidability_</code> <code>residual_pos</code>	<code>UgpLean.Gravity.PSCEpochSelection[†]</code>	zero sorry
<code>d_res_determines_omega_</code> <code>lambda</code>	<code>UgpLean.Gravity.PSCEpochSelection[†]</code>	zero sorry
<code>psp_from_psc_structure</code>	<code>UgpLean.Gravity.PSCEpochSelection[†]</code>	zero sorry
<code>psp_epoch_selection_master</code>	<code>UgpLean.Gravity.PSCEpochSelection[†]</code>	zero sorry
<code>omega_lambda_gte_approx</code>	<code>UgpLean.Gravity.PSCEpochSelection[†]</code>	zero sorry (inter- val)

Galois, Algebraic Structure, and Statistics

Theorem	Module	Status
<code>galois_z7_cyclic_order_6</code>	<code>UgpLean.Algebra.CyclotomicGalois[†]</code>	zero sorry
<code>galois_z7_cpt_generator</code>	<code>UgpLean.Algebra.CyclotomicGalois[†]</code>	zero sorry
<code>galsplit_q_z7_not_in_q_</code> <code>z120</code>	<code>UgpLean.Algebra.CyclotomicGalois[†]</code>	zero sorry
<code>fermionic_statistics_from_</code> <code>z7</code>	<code>UgpLean.Substrate.FermionicStatistics[†]</code>	zero sorry
<code>gte_baryon_number_</code> <code>topological_charge</code>	<code>UgpLean.Algebra.BaryonNumber</code>	zero sorry
<code>su3_cmca_master_bundle</code>	<code>UgpLean.Algebra.SU3GluonCount</code>	zero sorry

Geometric Structure and Continuum Limit

Theorem	Module	Status
rule110_gf7_vacuum_fixed_point_master	UgpLean.Spacetime.GF7VacuumFixedPoint [†]	zero sorry
gorard_ricci_flat_vacuum	UgpLean.Spacetime.GorardRicciFlatVacuum [†]	zero sorry
kappa_SD_eq_10_13	UgpLean.Spacetime.GorardKappa [†]	zero sorry
rule110_discrete_bianchi_identity	UgpLean.Spacetime.DiscreteBianchi [†]	zero sorry
gorard_chain_catAL_master_bundle	UgpLean.Spacetime.GorardMasterBundle [†]	zero sorry
W1_ge_of_lipschitz	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
W1_triangle	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
W1_eq_zero_iff	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
gluedCoupling_cost_le	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
W1_nonneg	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
W1_le_couplingCost	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
W1_attained	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
couplingCostSet_isCompact	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
couplingCostSet_eq_image	UgpLean.ContinuumLimit.WassersteinDistance [†]	zero sorry
classical_discrete_rg_entropy_rate	UgpLean.Gravity.CMBSpectralTilt [†]	zero sorry
weyl_algebraic_miracle	UgpLean.Gravity.CMBSpectralTilt [†]	zero sorry
weyl_log_deriv_at_planck	UgpLean.Gravity.CMBSpectralTilt [†]	zero sorry
beta_g_z2_formula	UgpLean.Gravity.CMBSpectralTilt [†]	zero sorry
n_s_formula	UgpLean.Gravity.CMBSpectralTilt [†]	zero sorry
n_s_less_than_one	UgpLean.Gravity.CMBSpectralTilt [†]	zero sorry

New Certifications: Vacuum Spatial GH Convergence

Theorem	Module	Status
vacuum_cmca_gh_converges_to_flat_space	UgpLean.Spacetime.VacuumGHConvergence (<i>ugp-lean</i>)	A _{Lean} , zero sorry
finGrid_family_totally_bounded	UgpLean.Spacetime. MatterGHPrecompactness (<i>ugp-lean</i>)	A _{Lean} , zero sorry
single_kink_gh_converges_to_flat	UgpLean.Spacetime. MatterGHPrecompactness (<i>ugp-lean</i>)	A _{Lean} , zero sorry

Open Lean Targets (Blocked on Mathlib Library Infrastructure)

Target	Module	Blocker
WaldEntropy (3 sorries)	UgpLean.Gravity.WaldEntropy [†]	Mathlib manifold integrals (Stage 3)
z2_eft_predicts_cmb_tilt (discharged $\mathbf{A}_{\text{Lean}}$)	UgpLean.Gravity.CMBSpectralTilt [†]	Definitionally discharged; physical EFE interpretation open (OQ-QG-1-Z ₂ -EFT)

[†] These modules were developed and certified as part of this paper. The module names and paths listed above reflect the canonical `ugp-lean` structure.

Analytically Derived Results, No Lean Cert Required ($\mathbf{A/D}$)

The following results carry full analytic derivations and computational confirmation ($\mathbf{A/D}$) but do not require Lean certification (the derivations are direct arithmetic consequences of established constants):

- **Level-2 field energy bound:** $V_{\max} = 2m_{\text{kink}}^2/49 = 3.44 \times 10^{-3} \text{ GeV}^4$ ($\mathbf{A/D}$ — from $\mathbb{Z}_7$ potential structure; verified by `strong_field_uv_bound.py`).
- **Kink Compton scale:** $a_{\text{kink}}/\ell_{\text{Pl}} = 4.21 \times 10^{19}$ ($\mathbf{A/D}$ — from BPS kink mass $m_{\text{kink}} = 8m_\tau/49$; verified by `strong_field_uv_bound.py`).
- **Kink Hawking critical mass:** $M_{\text{kink,crit}} = (49/8) M_{\text{crit}} = 2.04 \times 10^{40} \text{ MeV}$; Bose enhancement factor 5.64 at M_{crit} ($\mathbf{A/D}$ — from $\mathbb{Z}_7$ kink mass and standard Hawking formula; verified by `hawking_kink_emission.py`).

B Numerical Verification

The numerical results in this paper are verified by the scripts listed in Table 9. Scripts are located in `papers/44_quantum_gravity/scripts/`.

Table 9: Verification scripts for P44 results.

Script	Key result verified	Value
hawking_radiation_phimdl.py	$T_H = M_{\text{Pl}}^2/(8\pi M_{\text{BH}})$; M_{crit}	Eq. (10), (11)
greybody_factor_phimdl.py	Greybody suppression table	$M_{\text{BH}} > M_{\text{crit}}$
uv_finiteness_curved_ background.py	DeWitt-Schwinger classification; C_i values	$C_i \approx 41.76$
rt_wald_extension_proof.py	RT formula proof steps; $T_2 \Rightarrow T_3$	log 7 cancellation
flrw_gte_bounce.py	Bounce dynamics; $\dot{H}_B = 3/\pi^2$; T_{reh}	Eq. (15), (16)
mdl_initial_state_ scoring.py	MDL K-scores; $K_{\text{tot}} = \log_2 3$	Theorem 5.1
cyclotomic_z7_analysis.py	Galois group structure; CPT identification	Theorem 7.1
norfleet_tools_test.py	Bakry-Émery floor $\kappa_{\text{SD}} = 10/13$; Ihara-Bass $\rho(B) = 2.29$; $D_{\text{CF}} = 4.018$	§8.3
bianchi_extended_test.py	Extended Bianchi identity $k = 3 \dots 10$; max $ \sum \kappa < 10^{-14}$	rule110_discrete_ bianchi_identity
bakry_emery_saturation_ check.py	κ_{SD} floor saturation; $W = 8.80 \times 10^{-5}$	§8.3
strong_field_uv_bound.py	$V_{\text{max}} = 3.435 \times 10^{-3} \text{ GeV}^4$; $a_{\text{kink}} = 6.80 \times 10^{-16} \text{ m}$; EFT breakdown at $M = \pi/\sqrt{3} \approx 1.81$	§8.3.6, Eq. V_{max}
hawking_kink_emission.py	$M_{\text{kink,crit}} = (49/8)M_{\text{crit}} = 2.044 \times 10^{40} \text{ MeV}$; Planck factor 5.64 at M_{crit}	§3.2

Computational Tools

Numerical computations use Python 3.9+ with NumPy for linear algebra and array operations, and SciPy for integration and optimisation.

The machine-checked formalisation is written in Lean 4 with Mathlib. All theorems cited as **A**_{Lean} carry zero **sorry** and zero custom axioms beyond Lean’s standard foundations; named physics axioms are explicitly disclosed wherever used. The full theorem inventory is in Appendix A.

Full reproduction instructions are provided in REPRODUCE.md co-located with this paper’s source.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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